Degrees of Freedom

A portfolio's covariance matrix consists of stock variances along the diagonal terms and covariance terms on the off diagonals. The covariance matrix is a symmetric matrix since the covariance between stock A and stock B is identical to the covariance between stock B and stock A.

If a portfolio consists of n stocks the covariance matrix will be $n \times n$ and with n^2 total elements. The number of unique variance terms in the matrix is equal to the number of stocks, n. The number of covariance terms is equal to $(n^2 - n)$ and the number of unique covariance terms is $(n^2 - n)/2 = n \cdot (n-1)/2$.

The number of unique covariance parameters can also be determined from:

$$\textit{Unique Covariances} = \binom{n}{2} = \frac{n(n-1)}{2} \tag{6.30}$$

The number of total unique elements k in the $n \times n$ covariance matrix is equal to the total number of variances plus total number of unique covariances. This is:

$$k = n + \frac{n(n-1)}{2} = \frac{n \cdot (n+1)}{2} \tag{6.31}$$

In order to estimate these total parameters we need a large enough set of data observations to ensure that the number of degrees of freedom is at least positive (as a starting point). For example, consider a system of m equations and k variables. In order to determine a solution for each variable we need to have $m \geq k$ or $m - k \geq 0$. If $m < k$ then the set of equations is underdetermined and no unique solution exists. Meaning, we cannot solve the system of equations exactly.

The number of data points, d, that we have in our historical sample period of time is equal to $d = \mathrm{n} \cdot \mathrm{t}$ since we have one data point for each stock n. If there are t days in our historical period we will have $\mathrm{n} \cdot \mathrm{t}$ data points. Therefore, we need to ensure that the total number of data points, d, is greater than or equal to the number of unique parameters, k, in order to be able to solve for all the parameters in our covariance matrix. This is:

$$d \geq k$$

$$n \cdot t \geq \frac{n \cdot (n+1)}{2}$$

$$t \geq \frac{(n+1)}{2}$$